

Thomas Randall, PhD Computer Science

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SUMMARY:

Computer Science PhD with research focused on performance optimization and autotuning for High Performance Computing (HPC) workloads and applications, especially with GPUs

HIGHLIGHTED RESEARCH METRICS:

GPU Application Optimization

Improved cross-architecture scaling for Pascal- and Volta- NVIDIA GPUs, resulting in 472% higher kernel throughput while improving between-generation scaling by 197%.

Black Box Performance Modeling and Transfer Learning Optimization

Created a novel methodology for probabilistic black-box performance modeling that can be used to transfer automatic performance tuning between data- and system- scales, achieving up to 33.39x additional speedup over prior state-of-the-art performance tuning in a fraction of the evaluation cost.

Novel Hardware Effectiveness Study

Created an HPC-ready data harness to measure temperature sensors in conjunction with key performance metrics to determine the effectiveness of Single Phase Liquid Immersion Cooling apparatuses at maintaining HPC performance vs air-cooling techniques.

Deep Dive on LLM Capabilities

Proposed novel insights on the in-context learning phenomenon for LLMs which suggest that training data may inadvertently leak answers to models despite claims by model developers. This is further backed by low-level insights during the inference process that explain poor performance on useful tasks that these capabilities could serve in performance prediction and modeling domains.

EDUCATION:

PhD Computer Science

August 2019 – May 2025

Clemson University

Clemson, SC

Bachelor of Science in Computer Science

May 2019

Minor in Business Administration

Major GPA: 3.93/4.00

Clemson University

Clemson, SC

RELEVANT EXPERIENCE:

EXPERIENCE:

Volunteer Research Assistant

Fall 2025—Present

Clemson University

Clemson, SC

Graduate Research Assistant

Summer 2019—Spring 2025

Clemson University

Clemson, SC

Vice President

Fall 2024—Spring 2025

School of Computing Graduate Student Association

Clemson, SC

- Organized 3 professional workshops, 3 social events and a fundraiser
- Managed organization finances, plans, marketing and personnel

OMNI Internship Program, Graduate Student Researcher

Summer 2024

Oak Ridge National Laboratory

Oak Ridge, TN

Lead Student Volunteer, Guided Interest Group Leader

Fall 2024

SuperComputing'24

Atlanta, GA

- Jointly organized Reproducibility efforts with conference committee members
- Managed student volunteers to facilitate conference events
- Organized and engaged new students at SuperComputing through cohort events and discussions

Lead Student Volunteer, Guided Interest Group Leader, HPC Immersion Mentor

Fall 2023

SuperComputing'23

Denver, CO

- Jointly organized conference Panels and Birds of a Feather with conference committee members
- Managed 100+ student volunteers to facilitate conference events
- Organized and engaged new students at SuperComputing through cohort events and discussions

GIVENS Scholar, Graduate Student Researcher

Summer 2020, 2022, 2023

Argonne National Laboratory

Lemont, IL

Graduate Teaching Assistant

Fall 2019 — Spring 2023

Clemson University, School of Computing

Clemson, SC

- Prepared assignments, projects and homework for classes for 30-100+ students
- Aided students in office hours with questions about course materials and assignments
- Spring 2023: Developed new course for Clemson University with Assistant Professor Dr. Mert Pesé

Student Volunteer, Virtual Student Volunteer	Fall 2020 (Virtual), Fall 2021, Fall 2022
SuperComputing'20, SuperComputing'21, SuperComputing'22	Online; St. Louis, MO; Dallas, TX
- Moderate online discussions and forward questions to presenters - Enable communications between conference organizers and presenters	
Undergraduate Research Assistant	Summer 2018—Spring 2019
Clemson University	Clemson, SC
Laboratory Teaching Assistant	Spring 2017 — Spring 2019
Clemson University, School of Computing	Clemson, SC
- Prepared and introduced labs and projects for up to 30 students - Aided students with questions and problems during lab time and office hours	
Summer Staff Coordinator	Summer 2017
Isaiah 55 Deaf Ministries	La Feria, TX
Coordinated volunteer team construction projects as well as crafts and recreation with children in the community	

AWARDS:

Outstanding PhD Student in Computer Science	Spring 2025
ICS'21 Best Paper	Summer 2021
R.C. Edwards Graduate Fellow	Fall 2019—Spring 2022
Clemson School of Computing Outstanding Undergraduate Researcher	Spring 2019
Dupont Best Undergraduate Project	Spring 2019

INVITED TALKS AND FEATURED COVERAGE:

Oak Ridge National Lab Artificial Intelligence Seminar Series	Spring 2024
Invited Presenter	Oak Ridge, TN
Argonne National Lab Computer Science Seminar Series	Summer 2023
Invited Presenter	Lemont, IL
I Am HPC	Spring 2023
https://sc23.supercomputing.org/2023/05/hpc-on-the-rise-thomas-randall/	Online
Ask A Grad	Fall 2022
SoC-GSA & UPE	Clemson, SC